HelmholtzZentrum münchen
German Research Center for Environmental Health

Public bioinformatics resources for GWAS interpretation

HELMHOLTZ RESEARCH FOR GRAND CHALLENGES

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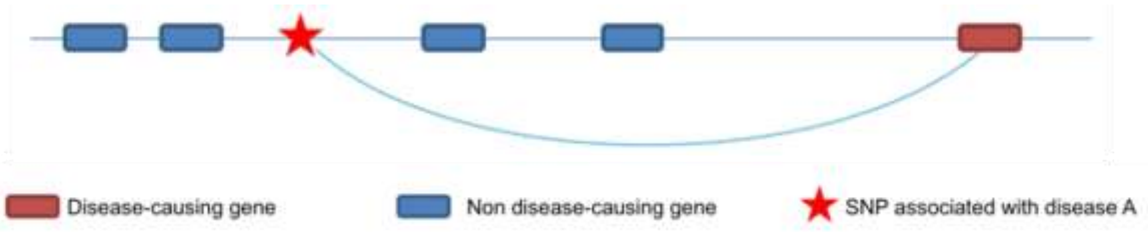
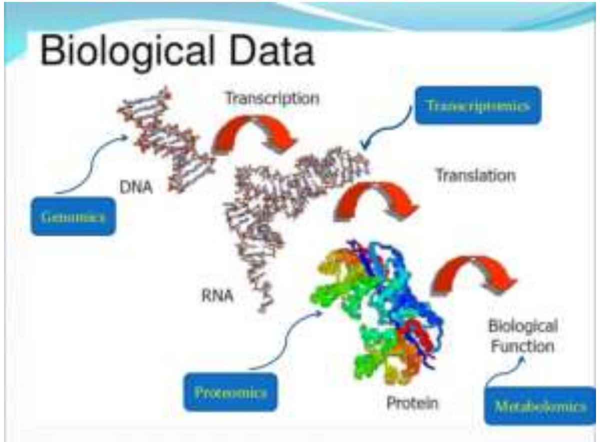
Institute of Translational Genomics

Motivation

- Greatest challenge in genetic association studies is to assign functionality (causality) to associated SNPs and genes.
- Predict the most likely genes and variants driving the phenotype associations.
- Need to integrate as much information as possible through genome annotation tools to drive the experimental validation.



What is Bioinformatics?



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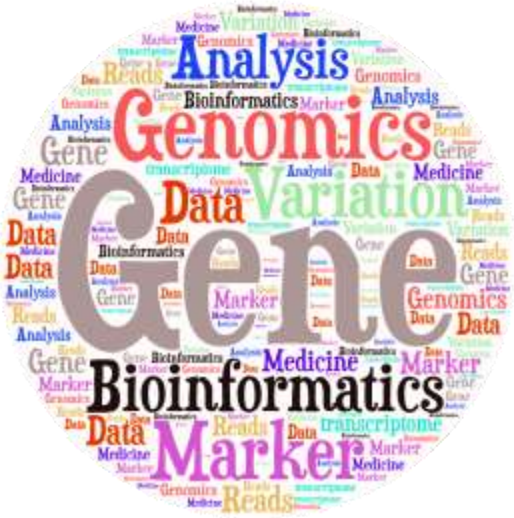
Computers aid to collect, intergrade and analyse biological data ->
Optimal biological interpretation

Goal of Bioinformatics

- Develop **databases** and **computational tools** to generate knowledge to better understanding a living cell and how it functions at molecular level.

Publicly available bioinformatics resources

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



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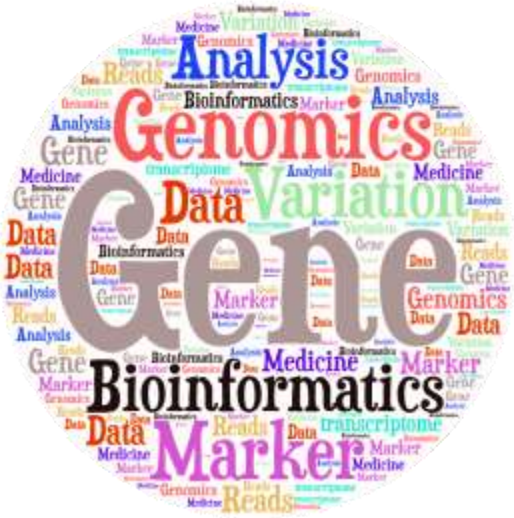
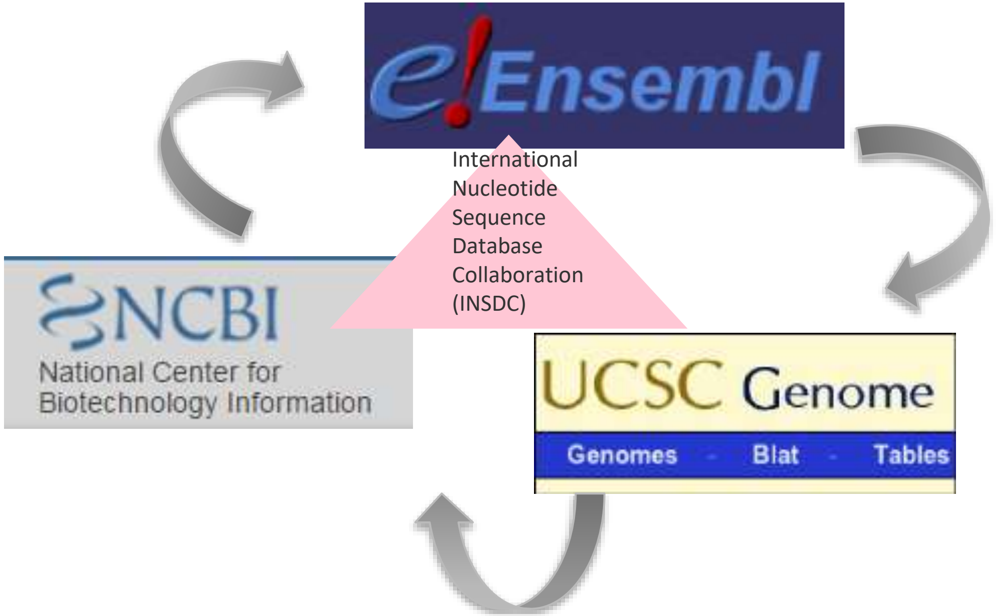


Genome Browsers

- Ensembl : http://www.ensembl.org
- UCSC: http://genome.ucsc.edu/
- NCBI: https://www.ncbi.nlm.nih.gov/

Genome Browsers

- Agreement: To begin bringing all genome assembly data into compliance moving forward



Ensembl … more than a browser!

- Ensembl is a joint project between the EBI (European Bioinformatics Institute) and the Wellcome Trust Sanger Institute
- **What can I do with Ensembl?**
  - Examine single nucleotide polymorphisms (SNPs) for a gene or chromosomal region.
  - View SNPs across strains (rat, mouse), populations (human) or breeds (dog).
  - View genes with annotations along the chromosome.
  - View alternative transcripts (i.e. splice variants) for a given gene.
  - Upload your own data.
  - Export sequence or create a table of gene information with BioMart.
  - Determine how your variants affect genes and transcripts using the Variant Effect Predictor.
  - Share Ensembl views with your colleagues and collaborators.

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  - Determine how your variants affect genes and transcripts using the Variant Effect Predictor.
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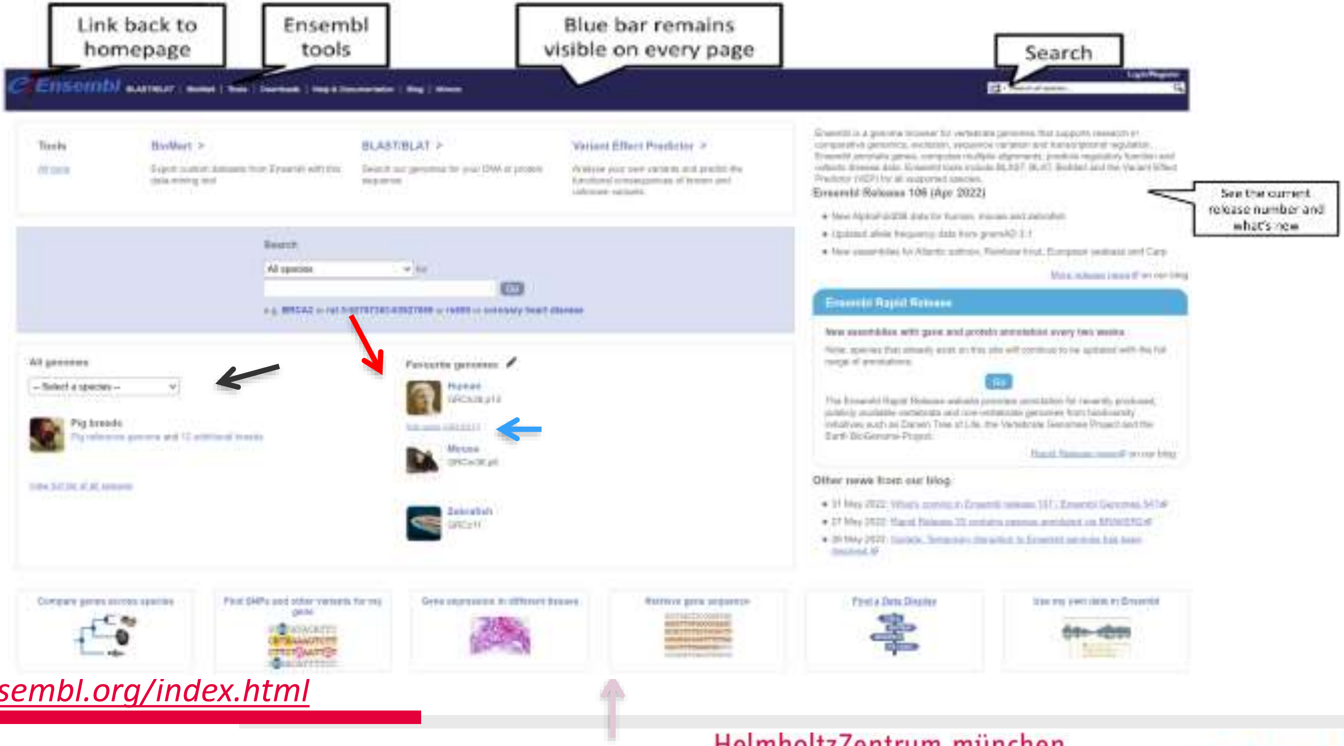
Ensembl … the front page

- It contains lots of information and links to help you navigate Ensembl.



Hint: https://www.ensembl.org/index.html

Ensembl … the front page … Exercise 1

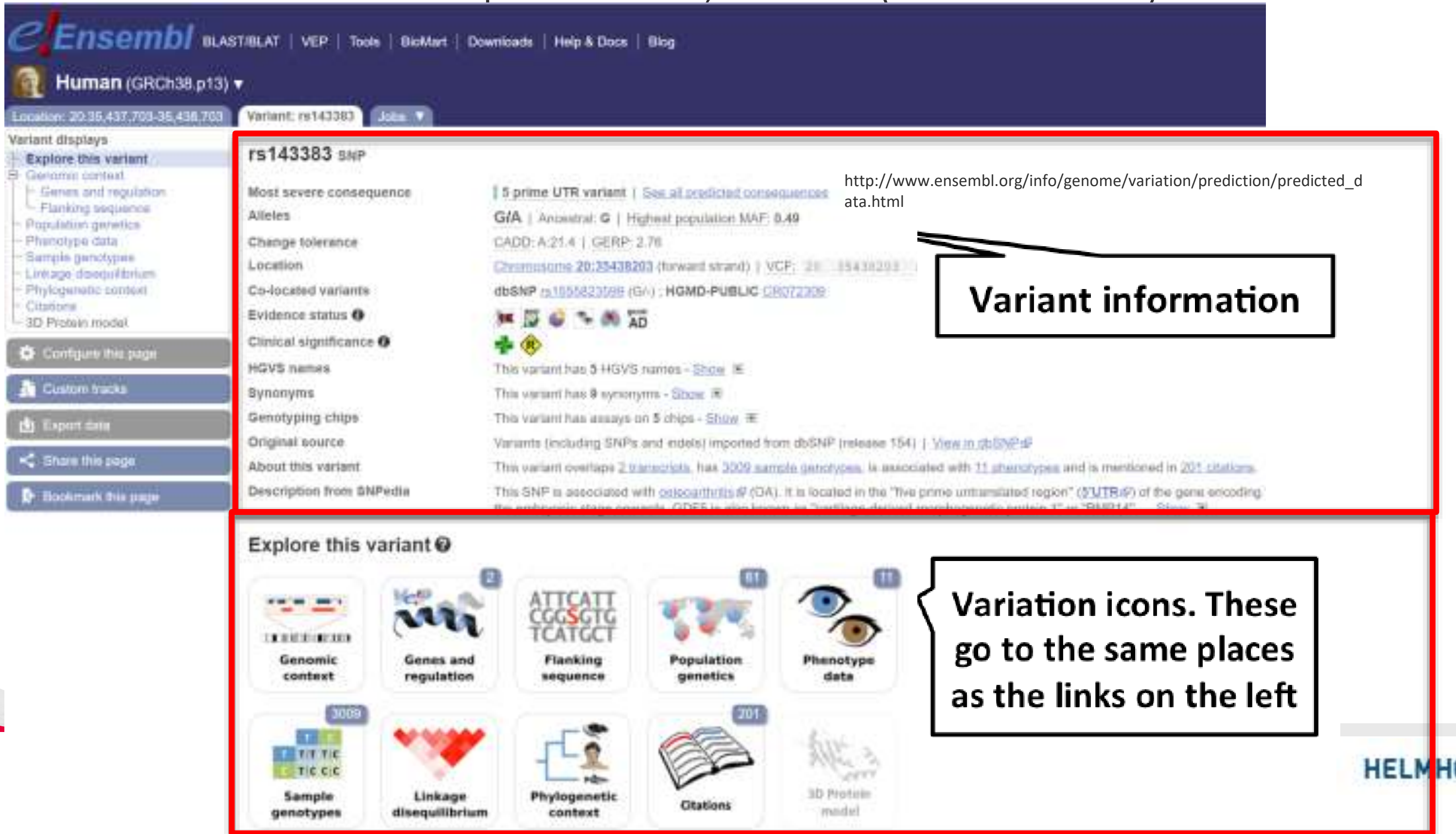
1. Go to the species homepage for **Giant Panda**. What is the name of the genome assembly for Panda?

2. How long is the Panda genome (in bp)? How many coding genes have been annotated?

3. Repeat for **Human**.

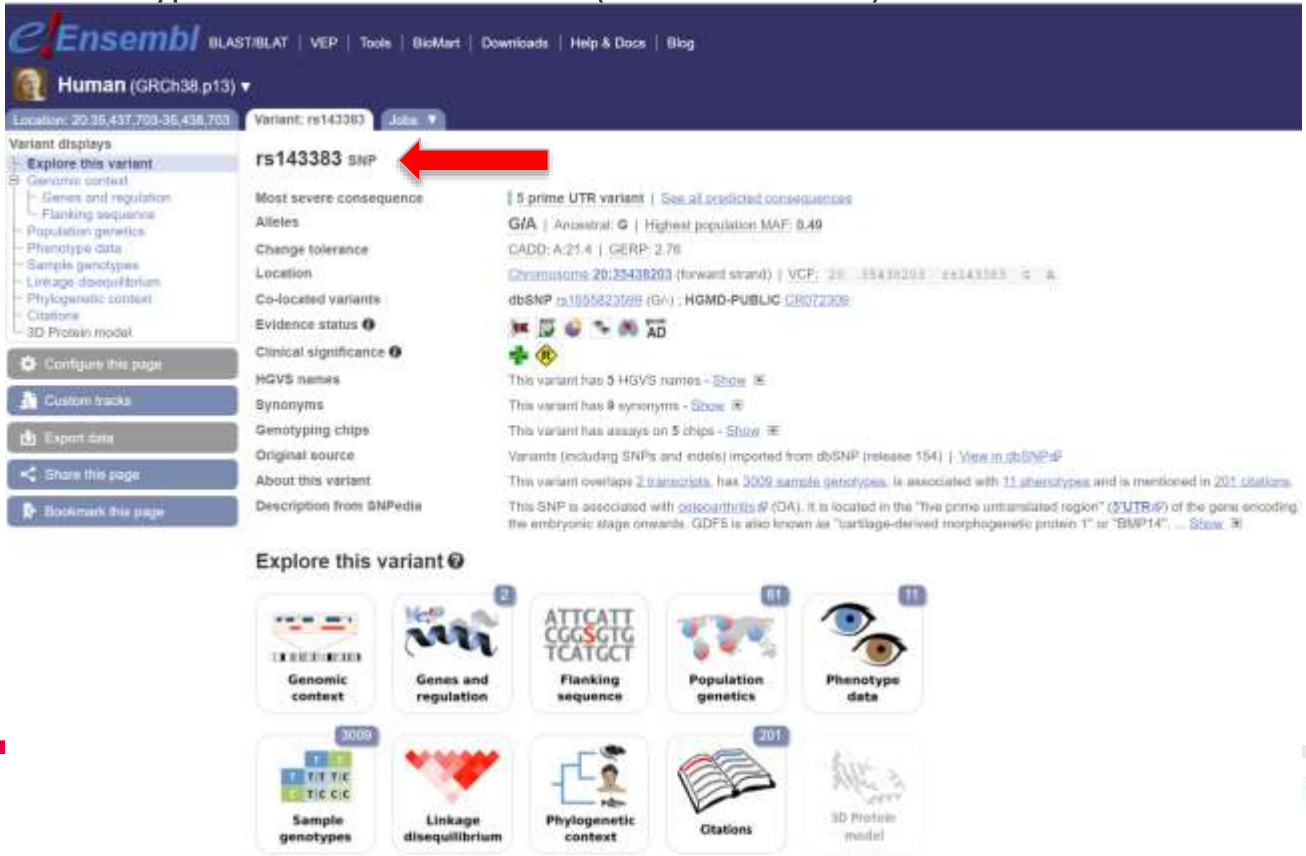
Ensembl … Exploring variants

- Let’s have a look at a specific variant, rs143383 (chr20:35438203).



Ensembl … Exploring variants …
Variant types

➢ What type of variants is rs143383 (chr20:35438203)?



Ensembl … Exploring variants …
Variant types

1. Small scale in one or few nucleotides of a gene

Type	Description	Example (Reference / Alternative)	
SNP	Single Nucleotide Polymorphism	Ref: ...TTGACGTA...	Alt: ...TTGGCGTA...
Insertion	Insertion of one or several nucleotides	Ref: ...TTGACGTA...	Alt: ...TTGATGCGTA...
Deletion	Deletion of one or several nucleotides	Ref: ...TTGACGTA...	Alt: ...TTGGTA...
Indel	An insertion and a deletion, affecting 2 or more nucleotides	Ref: ...TTGACGTA...	Alt: ...TTGGCTCGTA...
Substitution	A sequence alteration where the length of the change in the variant is the same as that of the reference.	Ref: ...TTGACGTA...	Alt: ...TTGTAGTA...

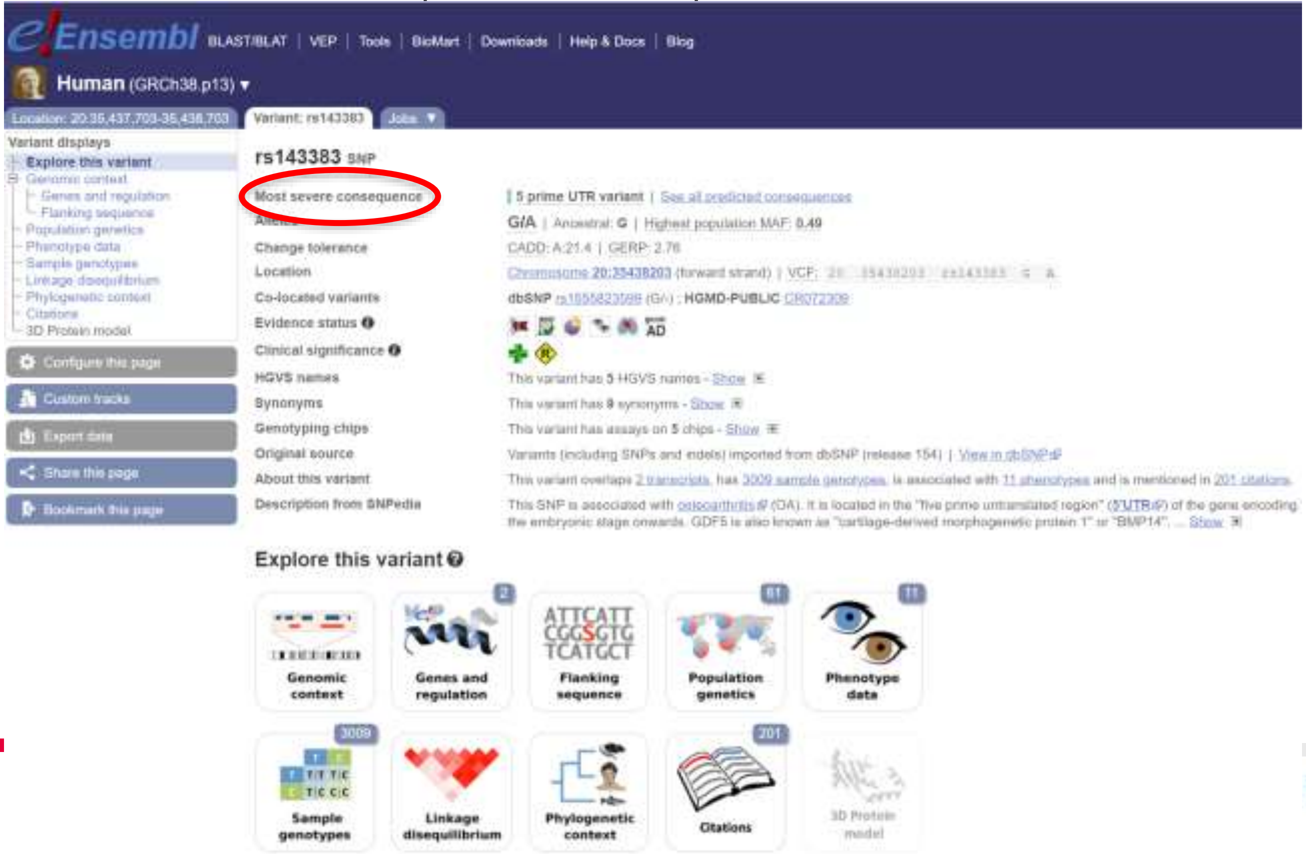
2. Large scale in chromosomal structure

CNV	Copy Number Variation: increases or decreases the copy number of a given region	Reference:	"Gain" of one copy: "Loss" of one copy:

deletion duplication insertion translocation

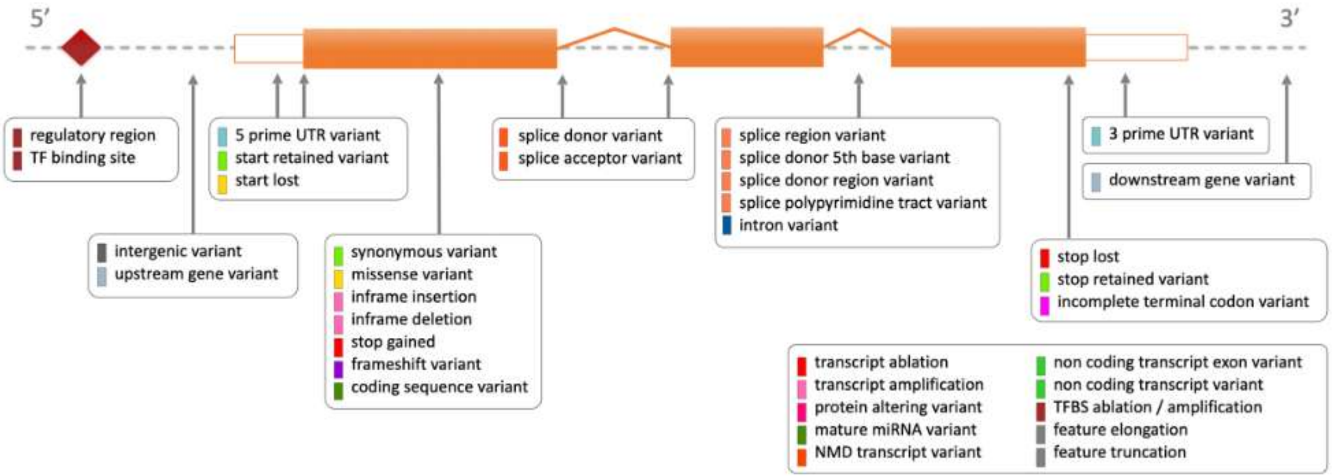
Ensembl … Exploring variants …
Calculated variant consequences

➢ Let’s have a look at the predicted consequences of the variant.



Ensembl … Exploring variants …
Calculated variant consequences

➢ A rule-based approach to predict the effects that each allele of the variant may have on the transcript.



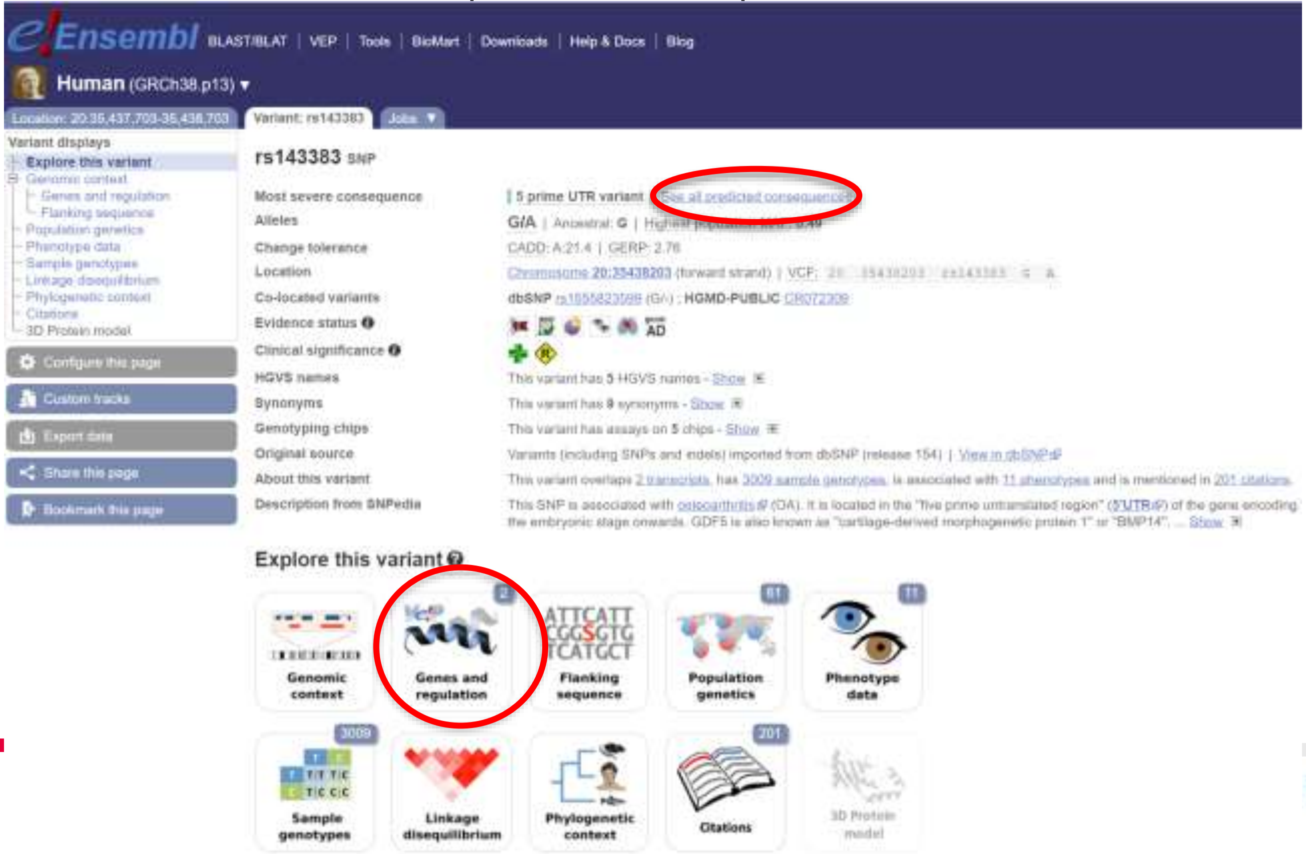
Ensembl … Exploring variants …
Calculated variant consequences

- Sequence Ontology (SO) consequence term :

http://www.ensembl.org/info/genome/variation/prediction/predicted_data.html

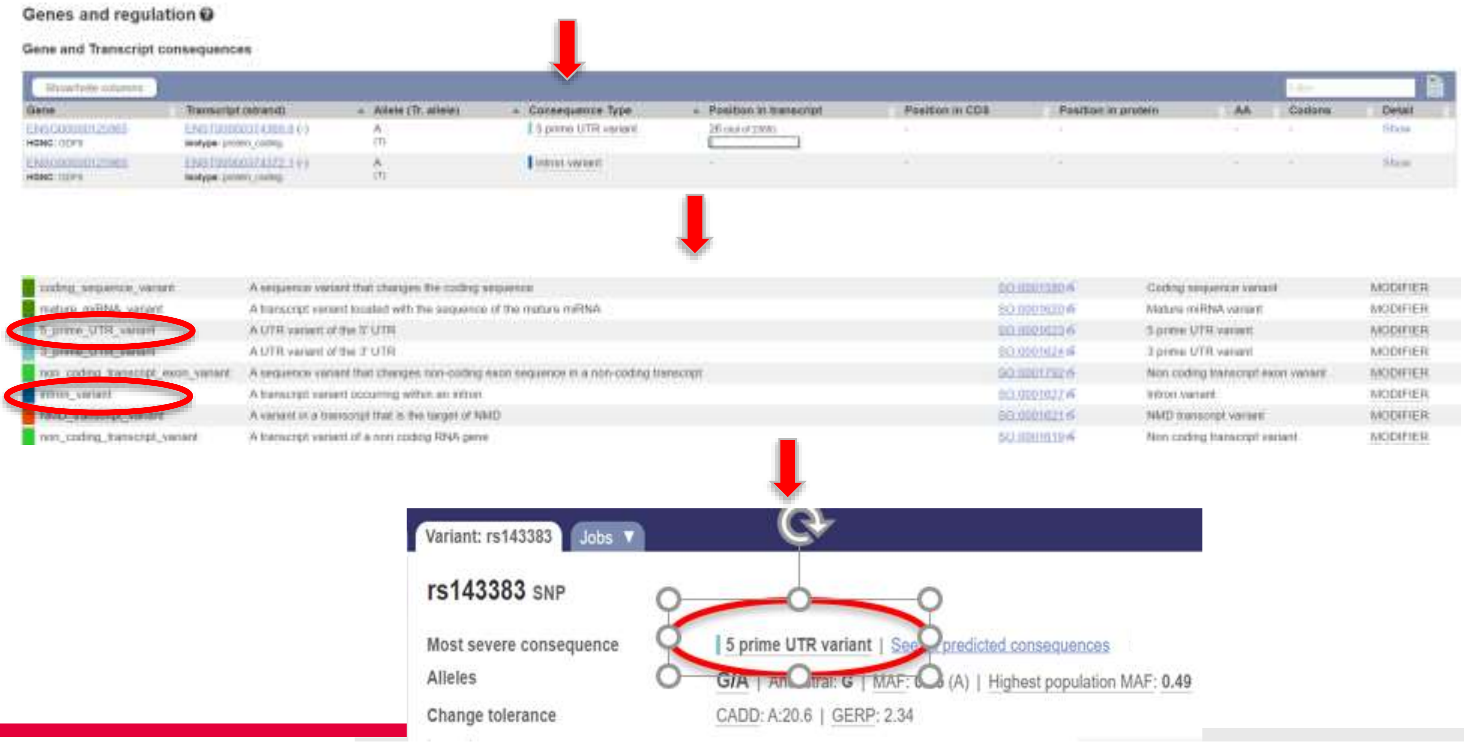
Ensembl … Exploring variants … Calculated variant consequences

➢ Let's have a look at all the predicted consequences of the variant.



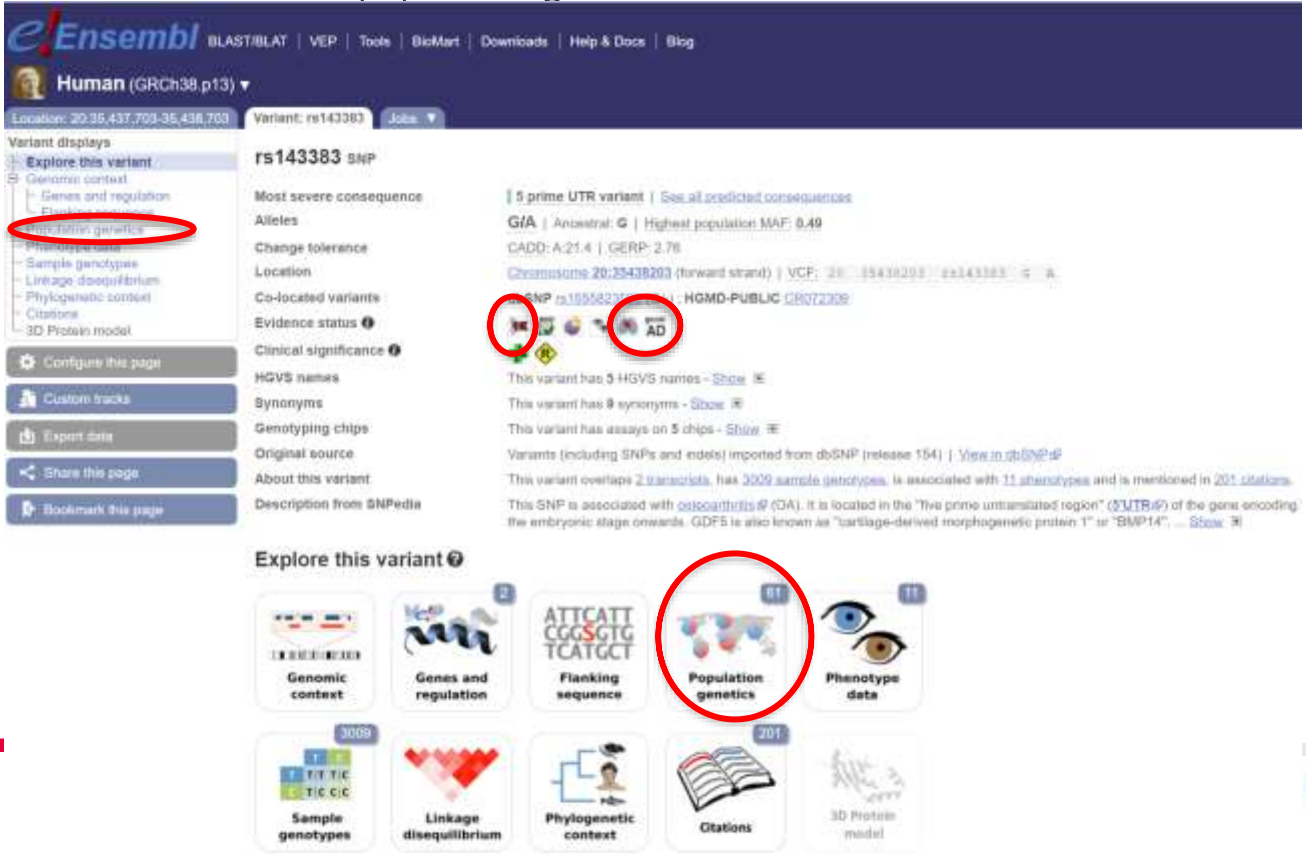
Ensembl … Exploring variants …
Calculated variant consequences

- ➢ This variant is found in 2 transcripts of the *GDF5* gene.



Ensembl … Exploring variants… Population genetics

➢ Let's have a look at population genetics.



Ensembl … Exploring variants …
Population genetics

- Frequencies are grouped by project for studies with multiple subpopulations.
- Pie charts can be displayed only for 1000 Genomes allele frequencies.

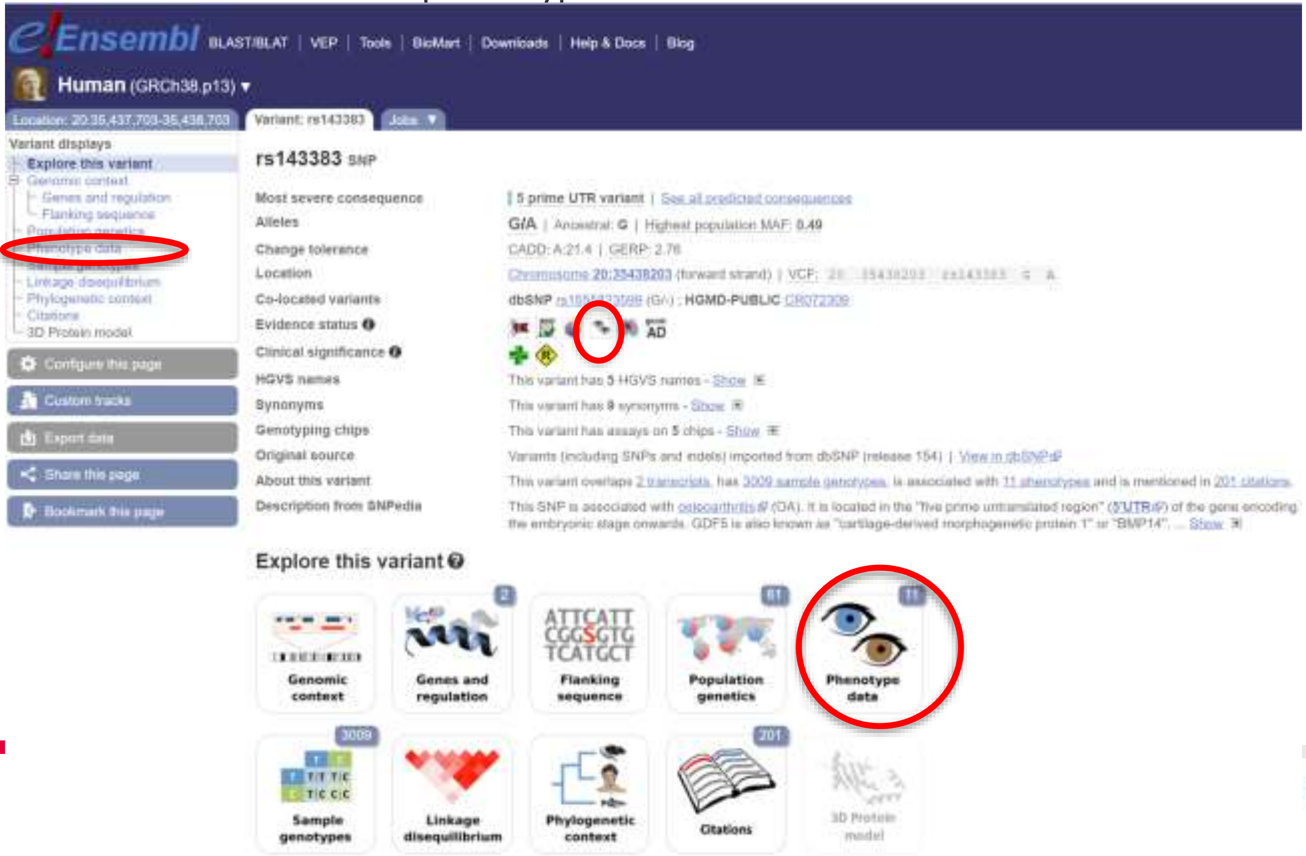
Tables of more detailed data

Expand subpopulations

Pie charts of allele frequencies

Ensembl … Exploring variants… Phenotype/Disease data

- ➢ Let's have a look at the phenotypes associated with this variant.



Ensembl … Exploring variants… Phenotype/Disease data

➢ This variant is associated with 15 phenotypes/diseases.

Phenotype Data

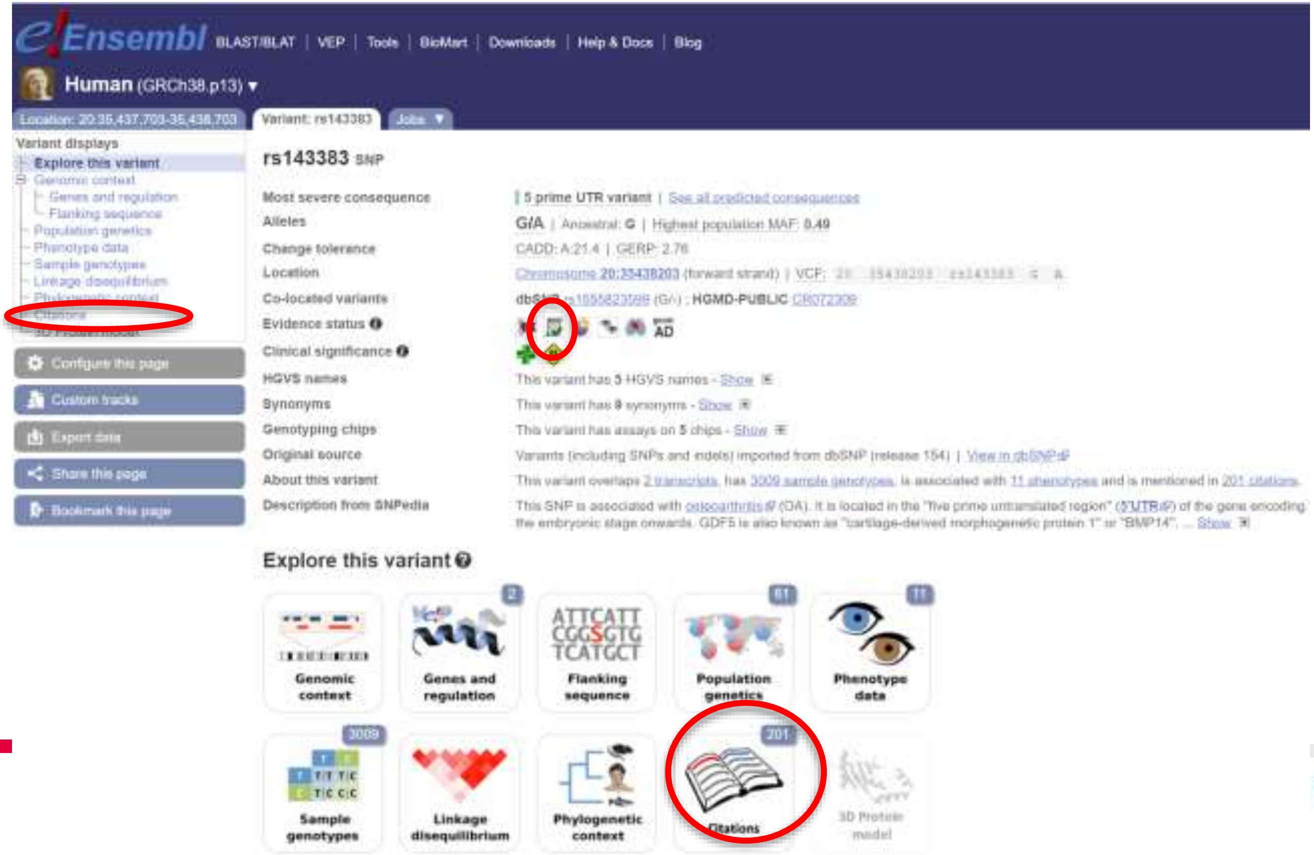
Significant association(s)

Show All entries | Show/hide columns

Phenotype, disease and trait	Source(s)	Mapped Terms	Ontology Accessions	Supporting evidence	External reference	Clinical significance	Reported gene(s)	Associated allele	Statistics
ACROMESOMELIC DYSPLASIA 2B	ClinVar [Illumina Laboratory ...]	-	Orphanet 53437	-	-		LOC109461475 GDF5	-	-
Acromesomelic dysplasia 2C, Hunter-Thompson type	ClinVar [Illumina Laboratory ...]	-	-	-	-		LOC109461475 GDF5	-	-
Brachydactyly	ClinVar [Illumina Laboratory ...]	Brachydactyly	HP 0001156, Orphanet 294937	-	-		LOC109461475 GDF5	-	-
ClinVar: phenotype not specified	ClinVar [GeneDx|Invitae]	-	-	-	-		LOC109461475 GDF5	-	-
Cortical surface area	NHGRI-EBI GWAS catalog	-	EFO 0010736	-	PMID 34560273	-	-	G	p-value: $1.00e^{-17}$
Grebe syndrome	ClinVar [Illumina Laboratory ...]	-	Orphanet 2098	-	-		LOC109461475 GDF5	-	-
Height	GIANT	body height	EFO 0004339	-	-	-	-	G	p-value: $2.81e^{-35}$
Height	NHGRI-EBI GWAS catalog	body height	EFO 0004339	-	PMID 34270796	-	GDF5	-	p-value: $1.00e^{-8}$ beta coefficient: 0.48 unit increase
Height	NHGRI-EBI GWAS catalog	body height	EFO 0004339	-	PMID 31562340	-	GDF5	-	p-value: $9.00e^{-78}$ beta coefficient: 0.06367727 unit increase
Knee osteoarthritis	NHGRI-EBI GWAS catalog	Knee osteoarthritis	EFO 0004616, HP 0005086	-	PMID 30664745	-	GDF5	-	p-value: $5.00e^{-7}$ odds ratio: 1.0495905
Knee osteoarthritis	NHGRI-EBI GWAS catalog	Knee osteoarthritis	EFO 0004616, HP 0005086	-	PMID 30374069	-	-	T	p-value: $8.00e^{-18}$ odds ratio: 1.09287
MULTIPLE SYNOSTOSES SYNDROME 2	ClinVar [Illumina Laboratory ...]	-	Orphanet 3237	-	-		LOC109461475 GDF5	-	-
Osteoarthritis of hip	ClinVar [OMIM]	-	-	PMID 17384641	MIM 601146		LOC109461475 GDF5	-	-
Vertex-wise cortical surface area	NHGRI-EBI GWAS catalog	-	EFO 0010736	-	PMID 34910505	-	-	G	p-value: $2.00e^{-16}$ beta coefficient: 0.25 z score increase
Whole brain free water diffusion multivariate analysis	NHGRI-EBI GWAS catalog	brain measurement	EFO 0004464	-	PMID 35505052	-	-	-	p-value: $3.00e^{-15}$

Ensembl … Exploring variants… Citations

- Let's have a look at the citations of the variant.



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Ensembl … Exploring variants … Citations

➢ This variant is mentioned in 232 citations.

Citations

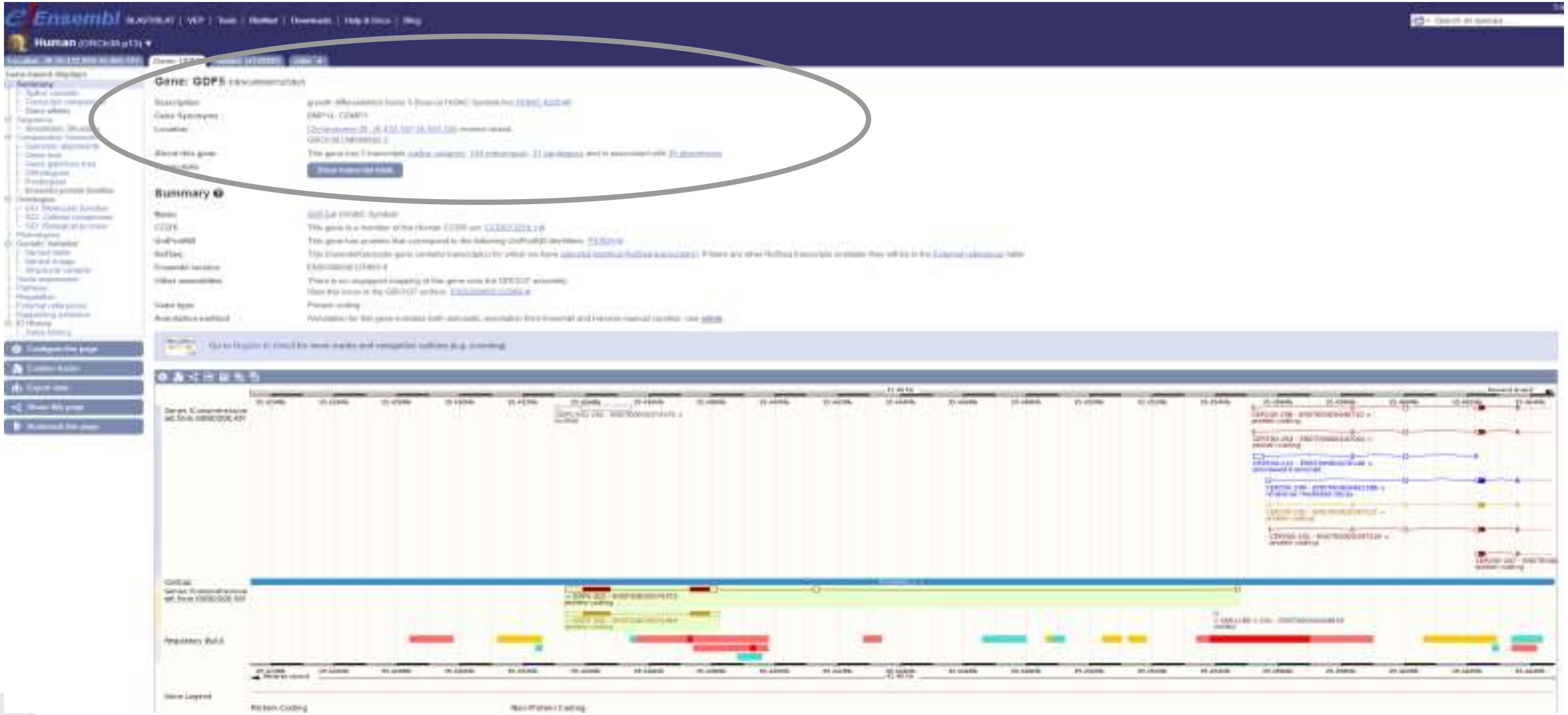
rs143383 is mentioned in the following publications

Show All entries | Show/hide columns

Year	PMID	Phenotype	Title	Author(s)	Full text	Citation source
2021	[illegible]		Identification of key genes in osteoarthritis using bioinformatics, principal component analysis and meta-analysis.	Sun X, Duan H, Xiao L, Yao S, et al	[illegible]	EPMC
2021	[illegible]		Current Evidence about Developmental Dysplasia of the Hip in Pregnancy.	Simionescu AA, Cirstoiu MM, Cirstoiu C, Stanescu AMA, et al	[illegible]	EPMC
2021	34210342		The effect of common variants in GDF5 gene on the susceptibility to chronic postsurgical pain.	Yan S, Nie H, Bu G, Yuan W, et al	[illegible]	EPMC
2021	[illegible]		The Mechanisms and Functions of GDF-5 in Intervertebral Disc Degeneration.	Guo S, Cui L, Xiao C, Wang C, et al	[illegible]	EPMC
2021	[illegible]		Growth differentiation factor 5 in cartilage and osteoarthritis: A possible therapeutic candidate.	Sun K, Guo J, Yao X, Guo Z, et al	[illegible]	EPMC
2021	[illegible]		Genetic Study of IL6, GDF5 and PAPPA2 in Association with Developmental Dysplasia of the Hip.	Harsanyi S, Zamborsky R, Krajciova L, Kokavec M, et al	[illegible]	EPMC
2021	[illegible]		A Case-Control Study of Major genetic pre-disposition risk alleles in developing DDD in the Northeast US population: Effects of Gene-Gene Interactions.	Vatchinov VI, Zhai SK, Hida T, Lacson R, et al		EPMC
2021	[illegible]		Frequency of Growth Differentiation Factor 5 rs143383 and asporin D-repeat polymorphisms in patients with hand and knee osteoarthritis in Kurdistan province, Iran.	Moghimi N, Naseeri S, Ghafouri F, Jalili A		EPMC
2020	31923126		A Meta-analysis Assessing the Association Between COL11A1 and GDF5 Genetic Variants and Intervertebral Disc Degeneration Susceptibility.	Wu F, Huang X, Zhang Z, Shao Z		dbSNP
2020	[illegible]		A novel variant near LSP1P5 is associated with knee osteoarthritis in the Chinese population.	Li Y, Liu F, Xu X, Zhang H, et al		EPMC,dbSNP
2020	[illegible]		Regulation of Gdf5 expression in joint remodelling, repair and osteoarthritis.	Kania K, Colella F, Riemen AHK, Wang H, et al	[illegible]	EPMC
2020	[illegible]		Association between growth differentiation factor 5 rs143383 genetic polymorphism and the risk of knee osteoarthritis among Caucasian but not Asian: a meta-analysis.	Peng L, Jin S, Lu J, Ouyang C, et al	[illegible]	EPMC
2020	[illegible]		Developmental Dysplasia of the Hip: A Review of Etiopathogenesis, Risk Factors, and Genetic Aspects.	Harsanyi S, Zamborsky R, Krajciova L, Kokavec M, et al	[illegible]	EPMC
2020	[illegible]		Can Implementation of Genetics and Pharmacogenomics Improve Treatment of Chronic Low Back Pain?	Surmanov V, Jovanovic F, Knezevic E, Candido KD, et al	[illegible]	EPMC
2020	[illegible]		Generation and characterization of human induced pluripotent stem cells (iPSCs) from hand osteoarthritis patient-derived fibroblasts.	Castro-ViÃ±uelas R, Sanjurjo-RodrÃguez C, PiÃ±eiro-Ramil M, [illegible]	[illegible]	EPMC

Ensembl … Exploring genes

➢ Let’s have a look at *GDF5* (growth differentiation factor 5).



Ensembl … Exploring variants … Exercise 2

Explore rs2075650 (build 37).

1. What type of variants is?
2. Where is this variant located (chr - position - strand)?
3. Are there big differences in allele frequencies between populations?
4. What is the ancestral allele frequency in the Japanese in Tokyo (JPT) population from the 1000G set?
5. What is the least frequent genotype for this variant in the KHV population from the 1000G set?
6. In how many transcripts is this variant found?
7. Are all transcripts from the same gene?
8. How many and which consequence type are predicted for this variant?
9. How many phenotypes are associated with this variant?
10. How many phenotypes are associated with the gene that this variant is located in?
11. How many publications refer to rs2075650?

Publicly available bioinformatics resources

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
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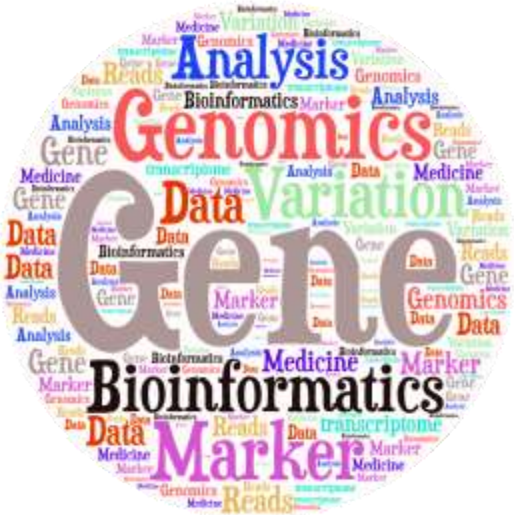
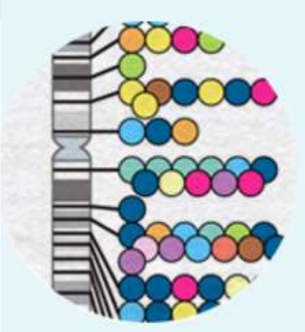


Human Traits & Diseases Databases

- GWAS catalog (https://www.ebi.ac.uk/gwas/)
- OMIM (http://omim.org/)
- Orphanet (http://www.orpha.net)



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Human Traits & Diseases Databases
GWAS catalog

- “GWAS catalog” is a catalog of published/unpublished genome-wide association studies.

2022

Human Traits & Diseases Databases
GWAS catalog

GWAS Catalog

The NHGRI-EBI Catalog of human genome-wide association studies

Search the catalog

Examples: breast carcinoma, rs7329174, Yao, 2q37.1, HBS1L, 6:16000000-25000000

- Let's have a look at a specific variant, rs143383.

Search results for *rs143383*

V rs143383

Location: 20:35438203 **Cytogenetic region:** 20q11.22 **Most severe consequence:** 5 prime utr variant **Mapped gene(s):** GDF5

Associations 6 Studies 6

G GDF5

Description: growth differentiation factor 5

Location: 20:35433347-35454746 **Cytogenetic region:** 20q11.22 **Biotype:** protein coding

Associations 92 Studies 67

Human Traits & Diseases Databases
GWAS catalog

V **rs143383**

Location: 20:35438203 **Cytogenetic region:**20q11.22 **Most severe consequence:** 5 prime utr variant **Mapped gene(s):** GDF5

Associations 10 Studies 10

Available data: Associations 10 | Studies 10 | Traits 5 | Linkage disequilibrium (LD) | Download Associations

Associations 10

Show 5 entries | Column visibility | Export | Clear search

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Background trait(s)	Study accession	Location
rs143383-G	2×10^{-19}	-	0.37	-	8.25 z score increase	-	GDF5	Vertex-wise cortical surface area	cortical surface area measurement	-	GCST90095130	20:35438203
rs143383-?	5×10^{-7}	-	NR	1.0495905	-	[1.03-1.07]	GDF5	Knee osteoarthritis	osteoarthritis, knee	-	GCST007090	20:35438203
rs143383-G	1×10^{-17}	-	0.3608	-	-	-	GDF5	Cortical surface area	cortical surface area measurement	-	GCST90091060	20:35438203
rs143383-T	5×10^{-18}	-	0.64	1.09287	-	[NR]	GDF5	Knee osteoarthritis	osteoarthritis, knee	-	GCST006925	20:35438203
rs143383-?	9×10^{-79}	-	NR	-	0.08367727 unit increase	[0.057-0.07]	GDF5	Height	body height	-	GCST000830	20:35438203

Showing 1 to 5 of 10 entries

‹ 1 2 ›

Human Traits & Diseases Databases
GWAS catalog

G **GDF5**

Description: growth differentiation factor 5

Location: 20:35433347-35454746 **Cytogenetic region:** 20q11.22 **Biotype:** protein coding

Associations 131 Studies 98

Human Traits & Diseases Databases …
Exercise 3

1. Try to find more information about rs2075650 and the gene that is located in GWAS catalogue and OMIM databases.

Publicly available bioinformatics resources

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- Nucleotide Sequence Databases
- Protein Sequence Databases
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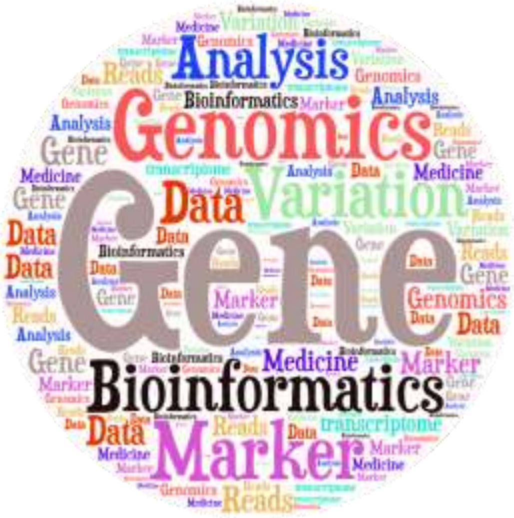
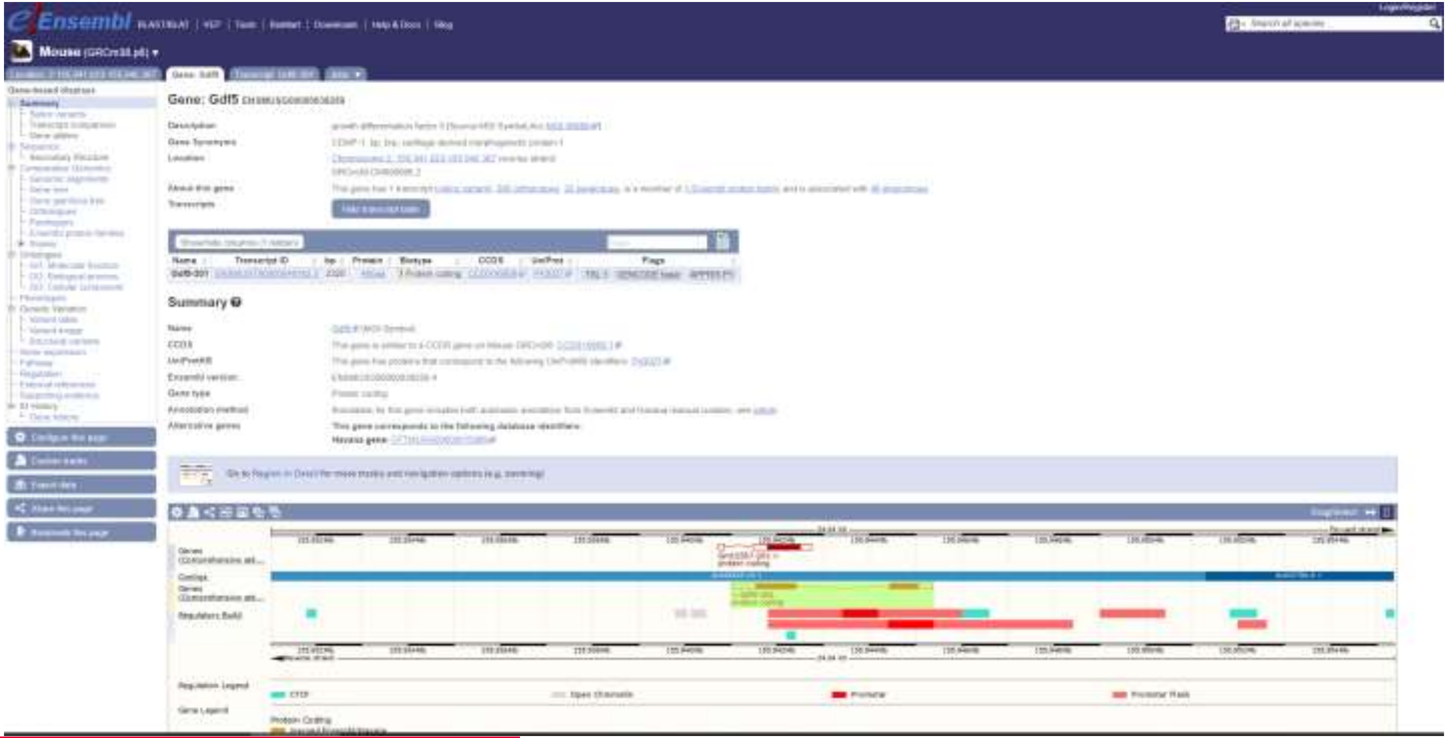


Databases of other Organisms

- Mouse: http://www.ensembl.org/Mus_musculus/Info/Index

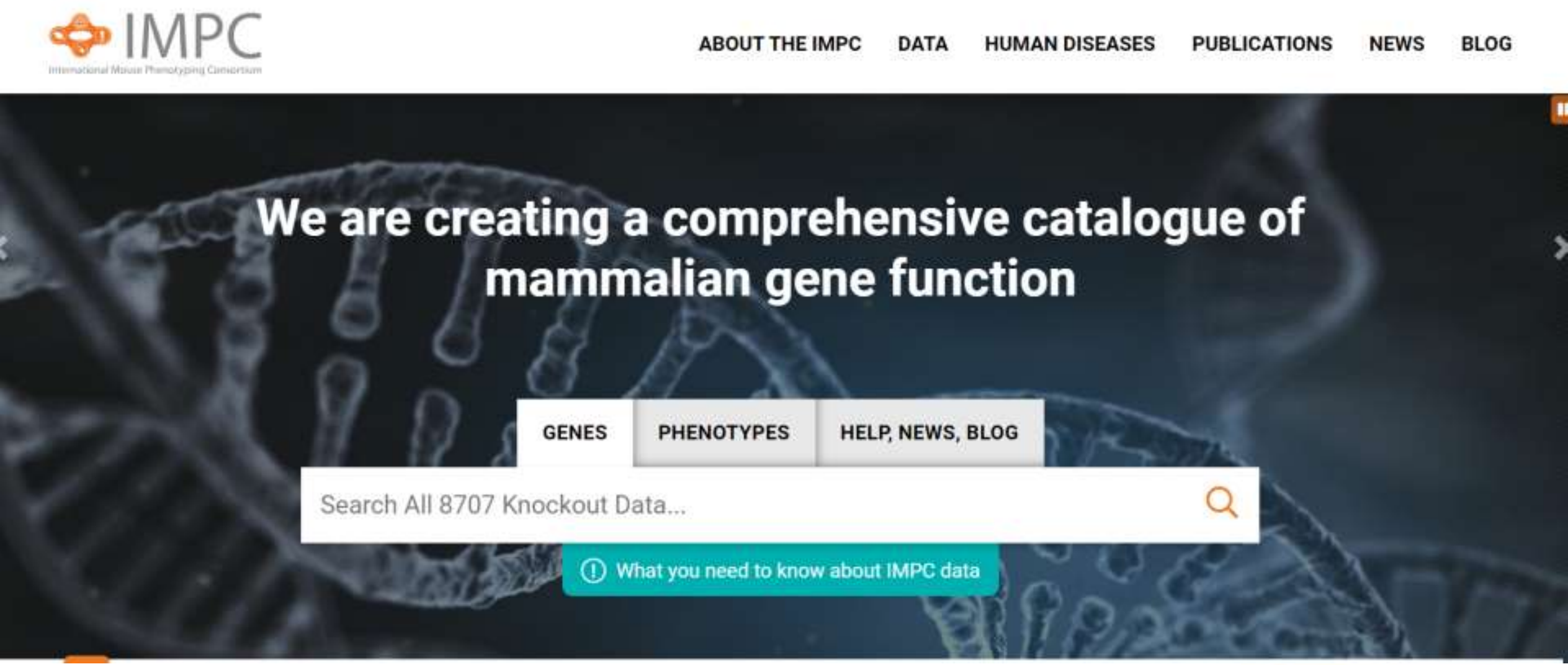


Databases of other Organisms

- Mouse: http://www.informatics.jax.org/

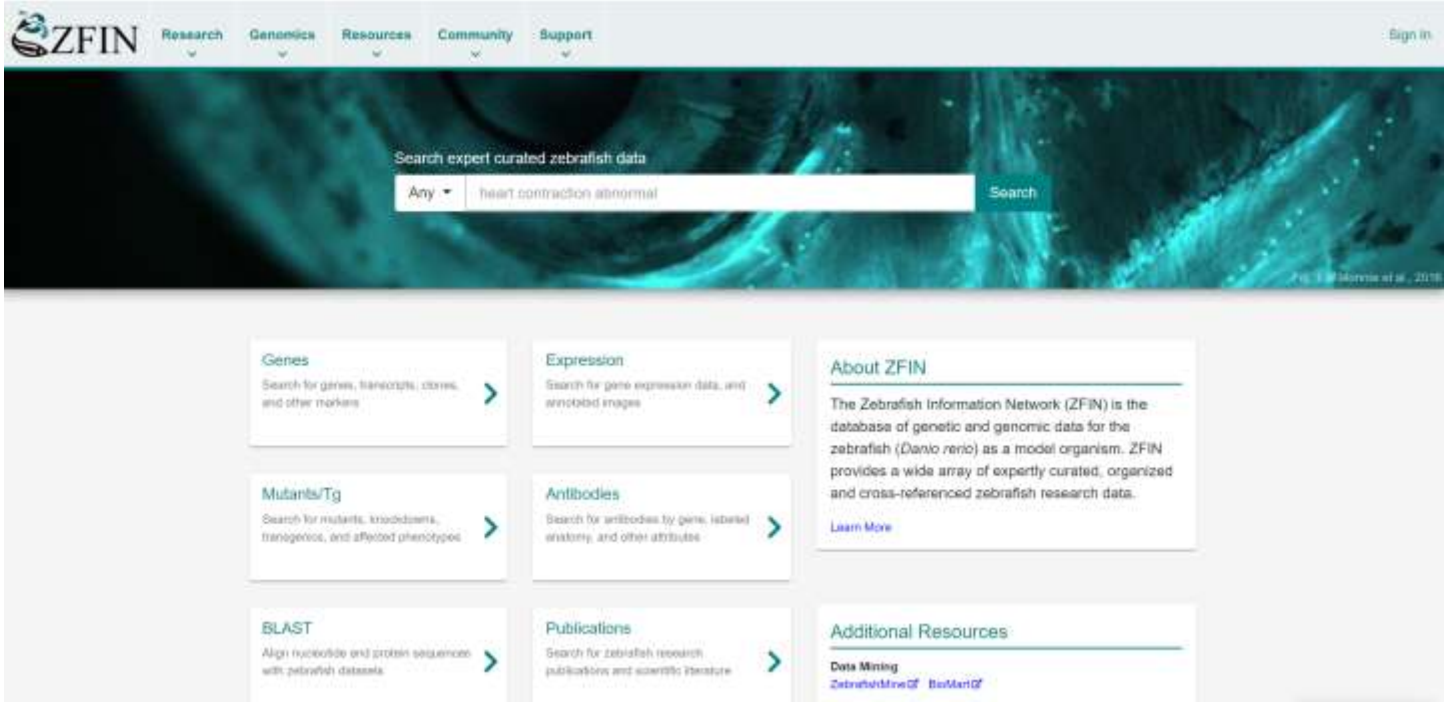
Databases of other Organisms

- Mouse: https://www.mousephenotype.org/



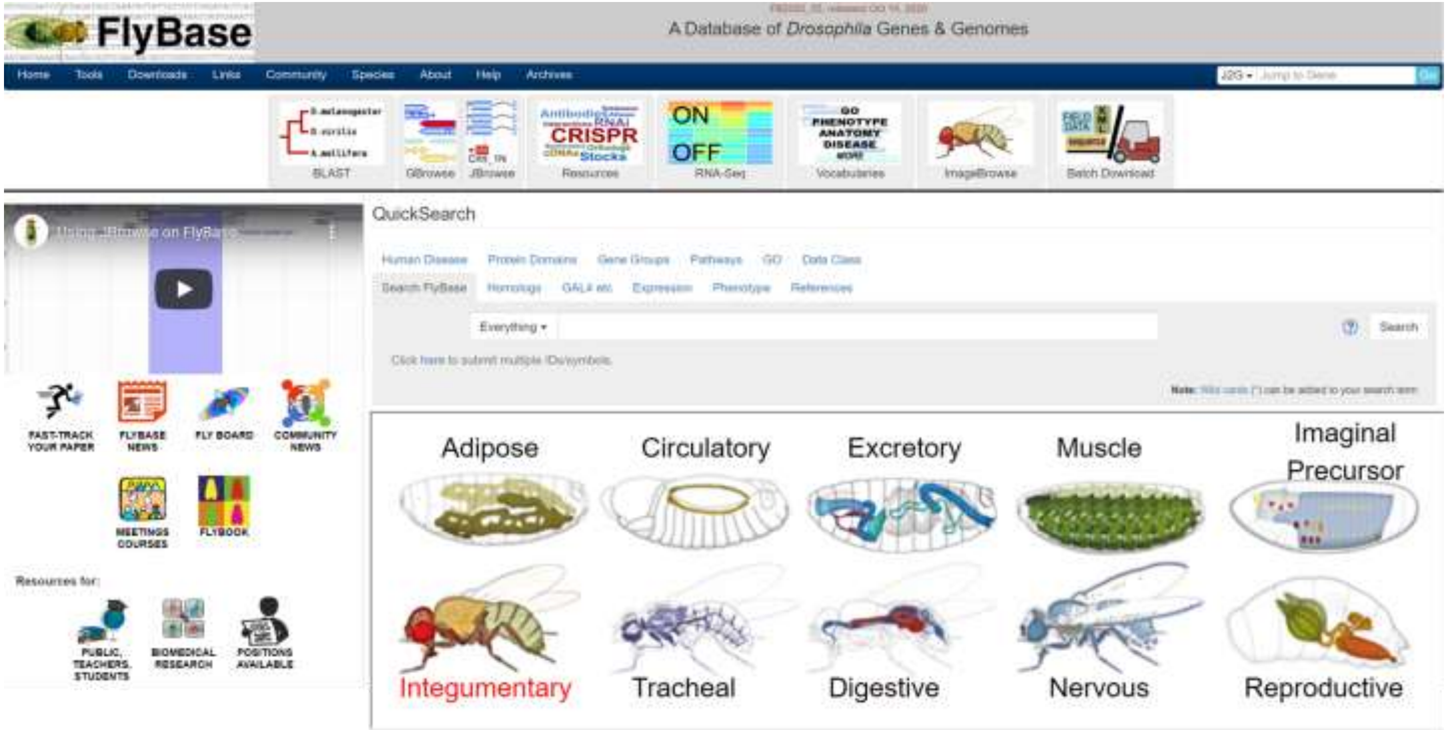
Databases of other Organisms

- Zebrafish: http://zfin.org/



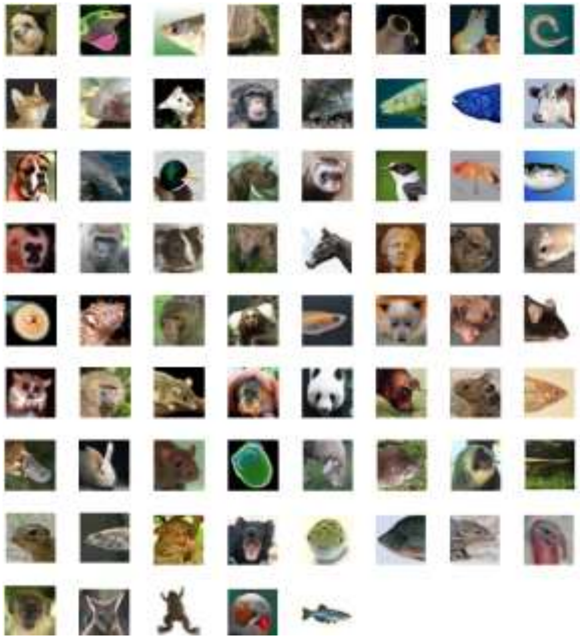
Databases of other Organisms

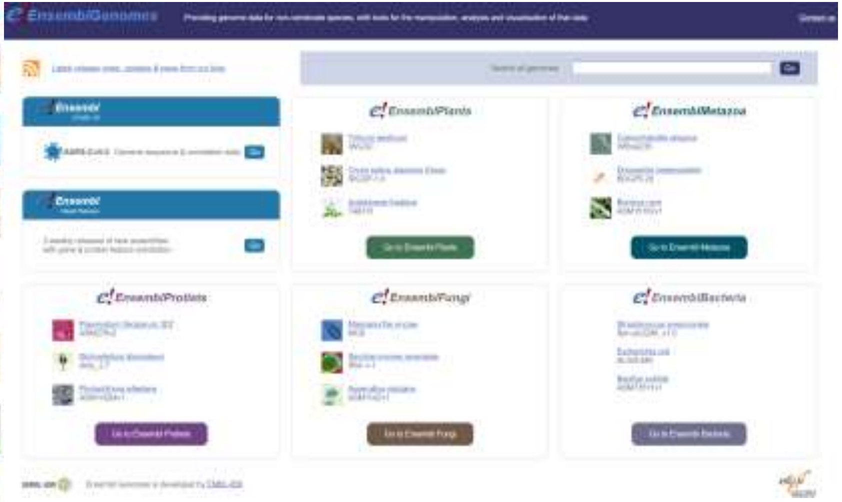
- Drosophila: http://flybase.org/



Databases of other Organisms

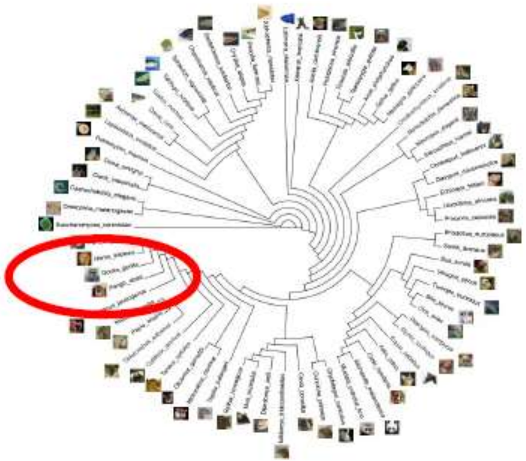
- Ensembl : >300 vertebrate species (https://www.ensembl.org/info/about/species.html) & >45000 genomes from non-vertebrate species (http://ensemblgenomes.org/)





Databases of other Organisms …
Exercise 4

- Explore rs2075650 in the databases of other organisms.
  - *Hint: you can search by rsID or (mapped) gene id or phenotype.*



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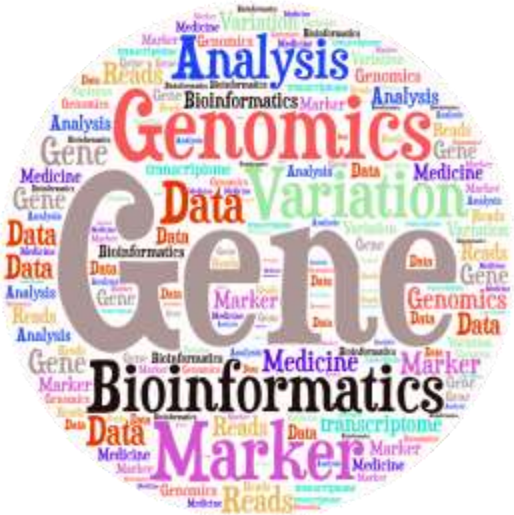
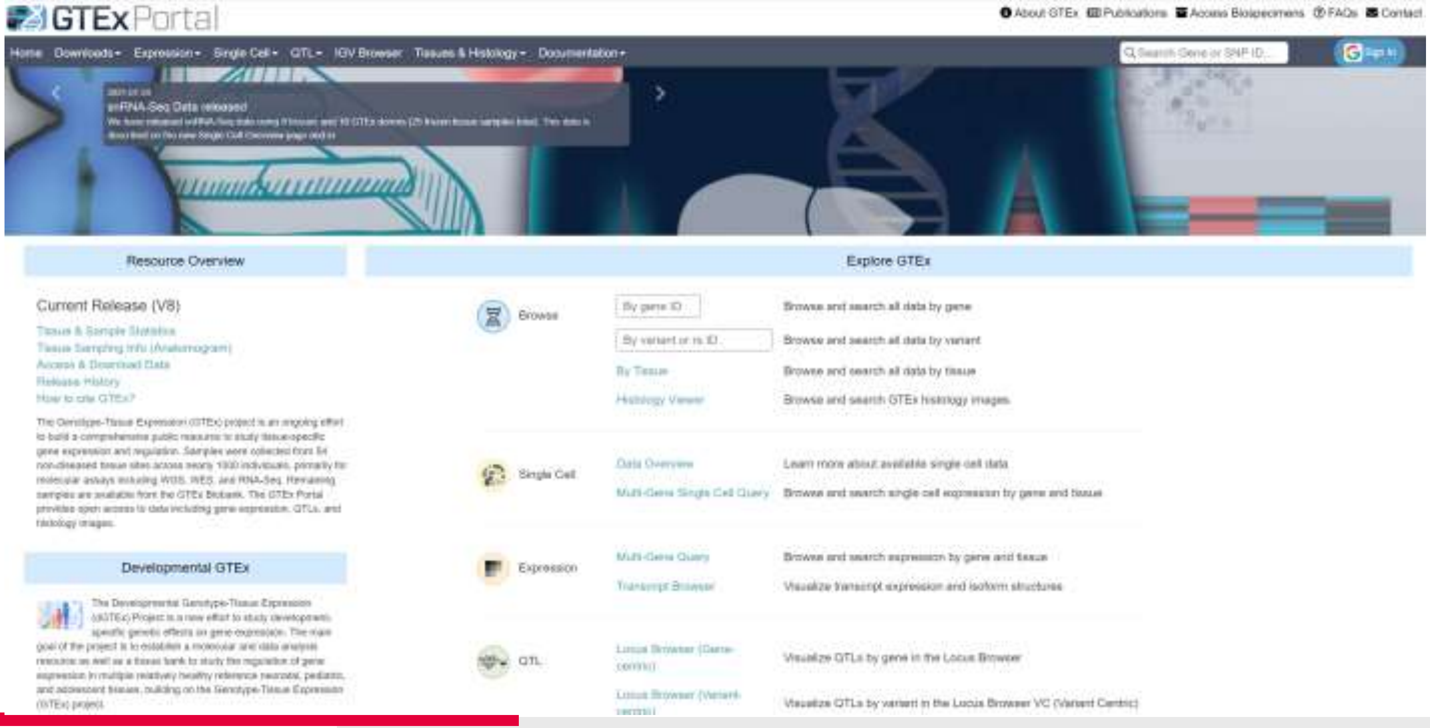


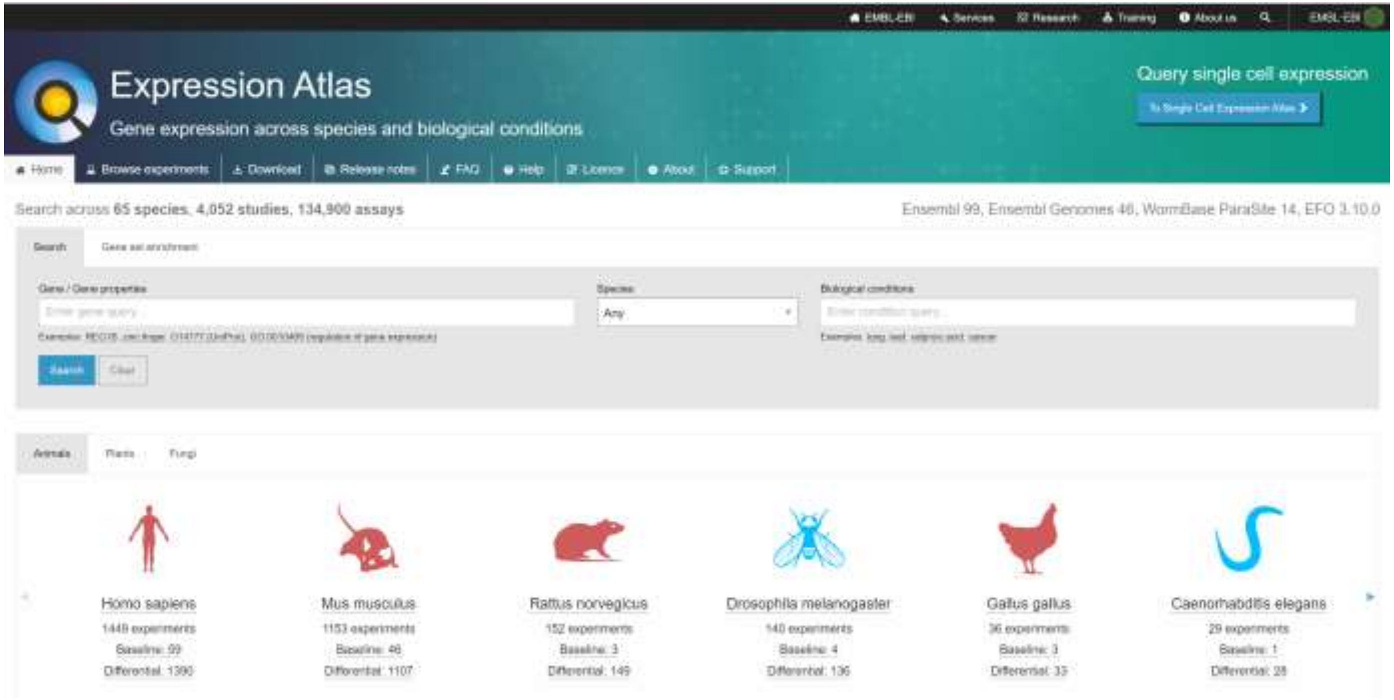
Gene Expression Databases

- Genotype-Tissue Expression (GTEx) : Correlations between genotype and tissue-specific gene expression (https://www.gtexportal.org/home/).



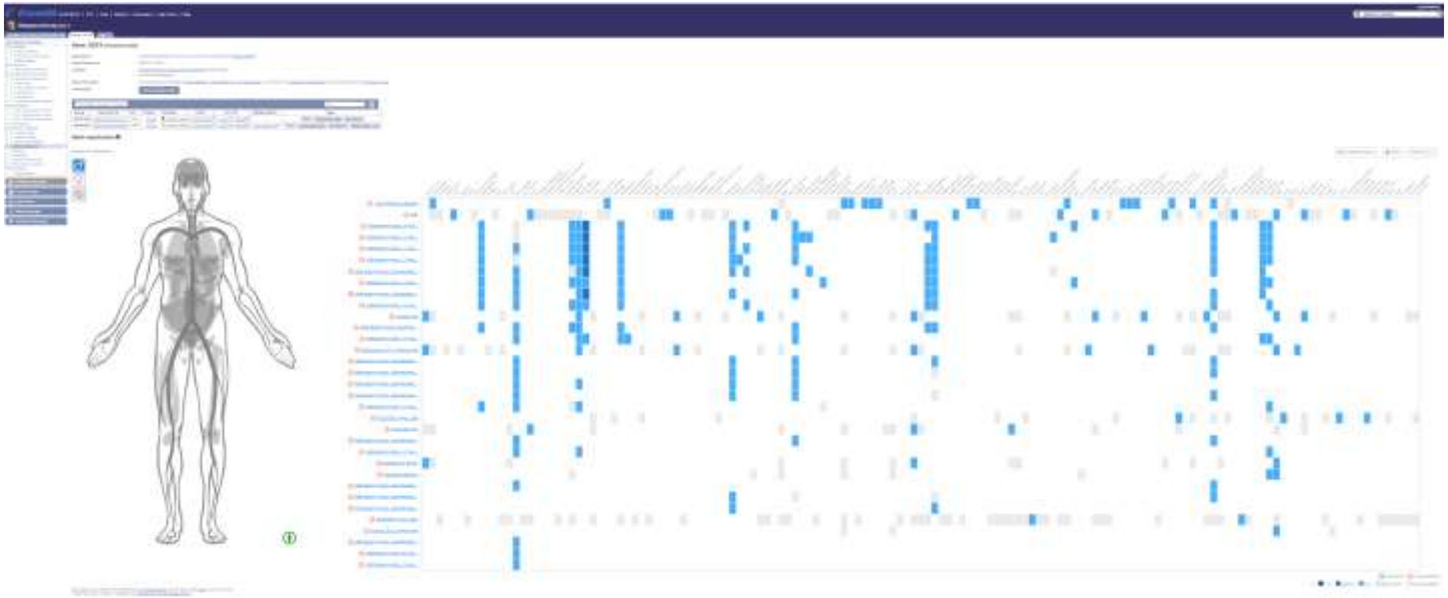
Gene Expression Databases

- Expression Atlas: Differential and Baseline Expression (http://www.ebi.ac.uk/gxa/home or through Ensembl).



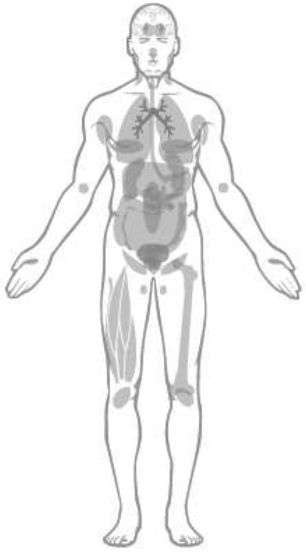
Gene Expression Databases

- Expression Atlas: Differential and Baseline Expression (http://www.ebi.ac.uk/gxa/home or through Ensembl).



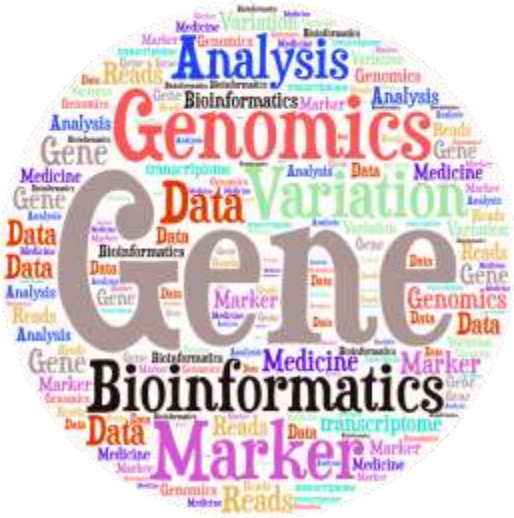
Gene Expression Databases …
Exercise 5

- Does rs2075650 affect the expression of the gene?
  - Hint: Use GTEx portal
- In which tissue *TOMM40* is expressed? Any connections with Alzheimer disease?
  - Hint: Use Expression Atlas



Publications Database

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



Publications Database

- PubMed (https://www.ncbi.nlm.nih.gov/pubmed) comprises over 34 million citations.
- How do I search PubMed?
  - Identify the key concepts for your search.
  - Enter the terms* in the search box.
    - Keywords
    - Author name
    - Journal name
    - .....
  - Suggestions will display as you type your search terms.
  - Click Search.

Publications Database

- Anatomy of the summary results:

1. Policy Issues in the Development and Adoption of Biomarkers for Molecularly Targeted Cancer Therapies: Workshop Summary.
   National Cancer Policy Forum, Board on Health Care Services, Institute of Medicine.
   Washington (DC): National Academies Press (US); 2015.
   PMID: **25855848** **Free Books & Documents**
   Similar articles

title

journal title abbreviation

authors

2. Four-wave mixing experiments with extreme ultraviolet transient gratings.
   Bencivenga F, Cucini R, Capotondi F, Battistoni A, Mincigrucci R, Giangrisostomi E, Gessini A, Manfredda M, Nikolov IP, Pedersoli E, Principi E, Svetina C, Parisse P, Casolari F, Danailov MB, Kiskinova M, Masciovecchio C.
   Nature. 2015 Apr 9;520(7546):205-8. doi: 10.1038/nature14341.
   PMID: **25855456**
   Similar articles

volume & issue

e-pagination

3. Molecular imaging of angiogenesis after myocardial infarction by (111)In-DTPA-cNGR and (99m)Tc-sestamibi dual-isotope myocardial SPECT.
   Hendrikx G, De Saint-Hubert M, Dijkgraaf I, Bauwens M, Douma K, Wierts R, Pooters I, Van den Akker NM, Hackeng TM, Post MJ, Mottaghy FM.
   EJNMMI Res. 2015 Jan 28;5:2. doi: 10.1186/s13550-015-0081-7. eCollection 2015.
   PMID: **25853008** **Free PMC Article**
   Similar articles

publicaton date

Publications Database … Exercise 6

- Try to find more information about rs2075650 and the gene that is located.
- Can you find a paper for which:
  - “Lopes” is one of the authors
  - Dealing with “osteoarthritis”
  - The name of the journal is “Lancet”

Pipeline

Dr Daniel Suveges
Senior Bioinformatician

Variant:

Ensembl:
- ✓ position,
- ✓ consequences (transcript and regulatory region),
- ✓ 1kG frequencies,
- ✓ GWAS signals around the variant,
- ✓ regulatory marks around the variant
- ✓ nearby genes.

ExAC: allele frequencies

GTEx: associated gene.

Pubmed: papers where the variant is cited.

Gene:

Ensembl:
- ✓ Position
- ✓ Associated phenotypes
- ✓ Gene ontology annotations

Uniprot:
- ✓ Function, Tissue, Localization, Disease

OMIM:
- ✓ gene annotation
- ✓ associated diseases

GWAS catalog:
- ✓ Associated variants and association details.

Gene Expression Atlas:
- ✓ Tissues in which expression of the gene was observed

GTEx:
- ✓ List of variants that affects the expression of the gene.

Mouse Genomics Informatics:
- ✓ Mouse phenotype of the gene.

Team144 pipeline

Variant details

General information:

rsID:	rs143384
Chromosome:start-end:	20:34025756-34025756
Allele string:	A/G
MAF:	0.438898
Consequence:	5_prime_UTR_variant
Variation type:	SNP
gerp (average gerp):	5.04 (3.497)
GWAVA score:	0.79
Ancestral allele:	G
Minor allele:	A
Synoyms:	rs431839, rs61433024, rs765106, rs3748435, rs17422934
Phenotypes (risk allele):	Height(A), Infant length(G), Height(G)

GWAS signals

GWAS signals within 1Mbp of the variant:

rsID	SNPID	Distance	Trait	p-value	PMID
rs17310467	chr20:33545616	-481kbp	Hemostatic factors and hematological phenotypes	4E-34	22443383
rs11906160	chr20:33565755	-461kbp	Anticoagulant levels	1E-6	22216198
rs1535466	chr20:33718706	-308kbp	Height	1.8E-29	25282103
rs2295888	chr20:33722863	-303kbp	Prothrombin time	5E-13	22703881
rs6120849	chr20:33730387	-296kbp	Protein C levels	7E-37	20802025
rs6088735	chr20:33745676	-281kbp	Hemostatic factors and hematological phenotypes	4E-34	22443383
rs6060278	chr20:33753262	-273kbp	Hemostatic factors and hematological phenotypes	4E-34	22443383
rs867186	chr20:33764554	-262kbp	Anticoagulant levels	4E-9	22216198
rs867186	chr20:33764554	-262kbp	Coagulation factor levels	6E-37	20231535
rs867186	chr20:33764554	-262kbp	D-dimer levels	4E-6	21502573
rs867186	chr20:33764554	-262kbp	Hemostatic factors and hematological phenotypes	2E-6	22443383
rs867186	chr20:33764554	-262kbp	Protein C levels	1E-64	25376901
rs11167260	chr20:33775200	-251kbp	Amyotrophic lateral sclerosis	4E-6	24529757

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Senior Bioinformatician

HelmholtzZentrum münchen
German Research Center for Environmental Health

Thank you!

HELMHOLTZ RESEARCH FOR GRAND CHALLENGES

What is a genome assembly?

Sequence reads

CGGCCTTTGGGCTCCGCCTTCAGCTCAAGA

CAGCTGTCCCAGATGAC ACTTAACTTCCCTCCCAGCTGTCC

GGGCTCCGCCTTCAGCTC AACTTCCCTCCCAGCT TCCCAGCTGTCCCAGATGACGCCATC

CGGCCTTTGGGCTCC CAGATGACGCC TCCGCCTTCAGCTCAAGACTTAACTTC

Match up overlaps

CGGCCTTTGGGCTCCGCCTTCAGCTCAAGA AACTTCCCTCCCAGCT CAGATGACGCC

TCCGCCTTCAGCTCAAGACTTAACTTC TCCCAGCTGTCCCAGATGACGCCAT

GGGCTCCGCCTTCAGCTC ACTTAACTTCCCTCCCAGCTGTCC

CGGCCTTTGGGCTCC CAGCTGTCCCAGATGAC

Genome assembly

CGGCCTTTGGGCTCCGCCTTCAGCTCAAGACTTAACTTCCCTCCCAGCTGTCCCAGATGACGCCAT

Ensembl … the front page … Exercise 1

1. Go to the species homepage for **Giant Panda**. What is the name of the genome assembly for Panda?
   - ➢ ASM200744v2, INSDC Assembly GCA_002007445.2, Apr 2020
2. How long is the Panda genome (in bp)? How many coding genes have been annotated?
   - ➢ 2,444,060,653 & 20,857
3. Repeat for **Human**.

➢ GRCh38.p13 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.28, Dec 2013

➢ 3,096,649,726

➢ 20,471 (incl 662 readthrough)

Ensembl … Exploring variants … Exercise 2

Explore rs2075650 (build 37).

1. What type of variants is? SNP
2. Where is this variant located (chr - position - strand)? Chromosome 19:44892362 (forward strand)
3. Are there big differences in allele frequencies between populations? Not really. Max 8% in some sub-African pop and 10% in som AMR sub-pop too.
4. What is the ancestral allele frequency in the Japanese in Tokyo (JPT) population from the 1000G set? 12%
5. What is the least frequent genotype for this variant in the KHV population from the 1000G set? G|G
6. In how many transcripts is this variant found? 5
7. Are all transcripts from the same gene? Yes, TOMM40
8. How many and which consequence type are predicted for this variant? 1, Intron
9. How many phenotypes are associated with this variant? 67
10. How many phenotypes are associated with the gene that this variant is located in? 1
11. How many publications refer to rs2075650? 313